EXP 5: COMPARATIVE ANALYSIS OF DIFFERENT TYPES OF PROMPTING PATTERNS AND EXPLAIN WITH VARIOUS TEST SCENARIOS

Aim:

To test and compare how different pattern models respond to various prompts (broad or unstructured) versus basic prompts (clearer and more refined) across multiple scenarios. Analyse the quality, accuracy, and depth of the generated responses.

Explanation:

Naïve Prompt:

A **naive prompt** is a straightforward question or command that does not include guidance, examples, or specific constraints, which may lead to less accurate or vague responses.

- **Example:**
  - **Naive prompt:**
    "Explain blockchain."

Basic Prompt:

A **basic prompt** is a clear and direct instruction that includes the main task and some guidance, but may still lack depth, examples, or advanced structure.

- **Example:**
  - **Basic prompt:**
    "Explain blockchain technology in simple terms."

SCENARIOS:

- Story writing
- Summarizing the article

- Answering the Factual Questions
- Giving Advice Over Driving

SCENE 1:

Naïve Prompt:

write a small story.

Response:

The old clock tower in the middle of the town had not worked for years. Its hands were stuck at 3:17, and no one really remembered when it had stopped. People passed by it every day without giving it much thought—except for a boy named Arjun.

Arjun was curious about everything. One evening, as the sky turned orange and the streets grew quiet, he decided to climb the narrow stairs inside the tower. Each step creaked as if it hadn't been used in ages. When he finally reached the top, he found the giant clock covered in dust.

As he looked closer, he noticed something unusual—a small glowing gear lying on the floor. It pulsed softly, like a heartbeat. Without thinking too much, Arjun picked it up and placed it into the clock's mechanism where a piece seemed to be missing.

The moment it clicked into place, the clock shuddered. The hands began to move again… slowly at first, then faster. Suddenly, everything around him changed. The dusty room became bright and new. The broken windows were whole again.

Arjun rushed outside and saw people dressed in old-fashioned clothes. Horse carts rolled along the street instead of bikes and cars. He realized, with a mix of fear and excitement, that the clock hadn't just started working—it had taken him back in time.

As the clock struck 3:18, Arjun smiled. He now understood that some things aren't broken—they're just waiting for the right person to bring them back to life.

Advanced or basic prompt:

Write a story which was happening in the future, where the world was suffering from the air pollution and focused on the hope of humanity in about 250-300 words.

Response:

In the year 2095, the world had changed beyond recognition. Thick layers of toxic smog covered the sky, turning daylight into a constant dull grey. People rarely stepped outside without heavy filtration masks, and children grew up without ever seeing a clear blue sky. Forests had disappeared, oceans were choked with waste, and fresh air had become a costly commodity stored in cylinders and sold to those who could afford it. Most cities were enclosed within massive glass domes, protecting the last pockets of breathable air.

Yet, even in this suffocating world, hope had not vanished. A group of young scientists from different parts of the globe came together with a shared dream—to bring back the Earth's lost breath. They worked tirelessly to create "aero-seeds," tiny floating devices designed to travel through the atmosphere. These seeds could absorb harmful pollutants and slowly release clean oxygen, mimicking the role of trees.

At first, people doubted them. The damage seemed too great, too permanent. But when the first batch of aero-seeds was released, something extraordinary happened. Weeks passed, and small patches of the sky began to clear. For the first time in decades, a faint shade of blue appeared above the horizon.

The change was slow, but it was real. Inspired by this progress, communities joined hands, planting new forests, reducing emissions, and protecting what little nature remained. Children began to learn about trees not just from books, but by seeing them grow.

Humanity realized that while it had nearly destroyed its world, it also possessed the courage and unity to restore it. And as the skies gradually brightened, so did the future—filled with the promise that even in the darkest times, hope could still rise.

SCENE 2:

Article:

India and the World Power

India is steadily emerging as a significant world power in the 21st century. With a large and youthful population, a rapidly growing economy, and strong advancements in technology, the country is gaining influence on the global stage. As one of the fastest-growing major economies, India plays an important role in global trade, innovation, and development.

In terms of geopolitics, India maintains strategic relationships with major countries while also promoting peace and cooperation. It is an active member of international organizations such as the United Nations and groups like BRICS and the G20, where it contributes to discussions on global issues including climate change, economic stability, and security.

India's strength also lies in its technological and scientific progress. The country has made remarkable achievements in space exploration, digital technology, and renewable energy. Initiatives like Digital India and advancements in startups have positioned India as a hub for innovation and entrepreneurship.

Moreover, India's cultural diversity and democratic values enhance its global image. As the world faces challenges such as climate change, economic inequality, and geopolitical tensions, India is increasingly seen as a responsible and influential player.

In conclusion, India's growing economic strength, strategic diplomacy, and technological progress are shaping its role as an emerging world power. With continued development and global cooperation, India has the potential to play a leading role in building a more balanced and sustainable world order.

Naïve Prompt:

Summarize the Article.

Response:

India is emerging as a major world power due to its fast-growing economy, young population, and technological advancements. It plays an important role in global organizations and maintains strong international relationships. With progress in innovation, space, and digital sectors, along with its democratic values, India is becoming an influential and responsible player in shaping the future world order.

Advanced or Basic Prompt:

summarize the article focused on India’s Strength and World's position changer relevant to the other country in about 250 words.

Response:

India’s rise as a global power is driven by a combination of economic strength, demographic advantage, technological progress, and strategic diplomacy. One of its greatest strengths is its large and youthful population, which provides a strong workforce and fuels innovation across industries. As one of the fastest-growing major economies, India has become a key player in global trade, attracting investments and expanding its influence in international markets.

Technological advancement is another major pillar of India’s strength. The country has made significant progress in areas such as digital infrastructure, space research, and renewable energy. Initiatives that promote startups and digital transformation have positioned India as a global innovation hub, enabling it to compete with developed nations.

On the global stage, India’s role is increasingly shaping the balance of power among nations. By actively participating in international organizations and forums, India contributes to important global discussions on climate change, economic policies, and security. Its approach to maintaining strategic partnerships with both developed and developing countries allows it to act as a bridge between different regions.

Compared to other countries, India is emerging as a strong alternative centre of growth and influence, especially in Asia. Its democratic values, economic resilience, and focus on sustainable development enhance its global image. As power dynamics shift in the world, India is not just growing internally but also redefining its position as a key contributor to global stability and progress.

SCENE 3:

Naïve Prompt:

Explain how the world created?

Response:

The creation of the world is explained mainly through science, especially the Big Bang Theory.

🌌 Scientific Explanation:

According to the Big Bang Theory, the universe began about 13.8 billion years ago from an extremely hot and dense point. Suddenly, it expanded rapidly—this event is called the Big Bang. As the universe expanded, it cooled down, allowing tiny particles to form atoms, mainly hydrogen and helium.

Over millions of years, gravity pulled these atoms together to form stars and galaxies. One of these galaxies is the Milky Way. Inside it, our Solar System

🌍 Formation of Earth:

The Earth formed as dust and rocks collided and stuck together. Over time, it cooled, forming land, oceans, and atmosphere. Eventually, simple life began to develop, leading to the diverse life we see today.

🌱 Other Perspectives:

Different cultures and religions also have their own creation stories, explaining the origin of the world in symbolic or spiritual ways.

✅ In short:

The world was created through natural processes starting from the Big Bang, followed by the formation of stars, planets, and finally Earth.

Advanced or Basic Prompt:

Describe How the world created by the theory of the big bang and how it evolved through the years by the recordings of the scientists in about 300 words.

Response:

The Origin and Evolution of the Universe

The origin of the universe is widely explained by the **Big Bang Theory**, a scientific theory stating that the universe began about **13.8 billion years ago** from an extremely hot, dense point. According to scientists, this tiny point suddenly

expanded in a massive event called the Big Bang. This expansion marked the beginning of space, time, matter, and energy.

In the first few minutes after the Big Bang, the universe cooled enough for basic particles such as protons, neutrons, and electrons to form. These particles combined to create the first atoms, mainly hydrogen and helium. Over millions of years, gravity pulled these atoms together, forming the first stars and galaxies. One of these galaxies is the **Milky Way**, which contains billions of stars.

Scientists have gathered evidence supporting the Big Bang through various discoveries. In 1929, astronomer **Edwin Hubble** observed that galaxies were moving away from each other, showing that the universe is expanding. Another key discovery was the **Cosmic Microwave Background Radiation**, detected in 1965 by **Arno Penzias** and **Robert Wilson**. This faint radiation is considered leftover heat from the early universe and provides strong evidence for the Big Bang theory.

Over billions of years, galaxies formed, stars were born and died, and heavy elements were created. About **4.6 billion years ago**, our **Solar System** formed from a cloud of gas and dust. Within it, the **Earth** developed conditions suitable for life.

Today, scientists continue studying the universe using powerful telescopes and satellites. Their observations show that the universe is still expanding and evolving, helping humanity better understand its cosmic origins.

SCENE 4:

Naïve Prompt:

Give some advice over how to drive a car.

Response:

▯ 🚙 **Stay focused** – Always keep your attention on the road and avoid distractions like mobile phones.

▯ 👀 **Be aware of surroundings** – Watch other vehicles, pedestrians, and road signs carefully.

⍰ ⚖️ **Control your speed** – Drive at a safe speed and slow down in traffic or bad weather.

⍰ 🛑 **Maintain safe distance** – Keep enough space between your car and the one ahead.

⍰ 🤝 **Follow rules and stay calm** – Obey traffic rules and don't get angry or impatient while driving.

Advanced or Basic Prompt:

Give Some confident advice to the person who has started to learn Driving and tries to drive in the highways which helps him to drive carefully.

Response:

Starting highway driving can feel intimidating—but with the right approach, you can handle it confidently and safely. Here's some practical advice:

🚚 Build Confidence Gradually

Don't rush into heavy traffic. Start on less busy highways and practice during off-peak hours to get comfortable with speed and lane discipline.

👀 Look Far Ahead

On highways, things happen fast. Keep your eyes far ahead (not just the car in front) so you can anticipate braking, lane changes, or obstacles early.

⚖️ Maintain a Steady Speed

Avoid sudden acceleration or braking. Try to match the flow of traffic while staying within speed limits.

🛣️ Stay in the Correct Lane

Use the left lanes for overtaking and stick to the middle or right lanes when driving normally (depending on local rules). Don't switch lanes unnecessarily.

🪞 Check Mirrors Frequently

Always check your mirrors before changing lanes. Use indicators early so other drivers understand your movement.

🚫 Keep a Safe Following Distance

At higher speeds, maintain more distance than in city driving. This gives you enough time to react safely.

🤝 Stay Calm and Don't Panic

If someone honks or overtakes aggressively, stay calm. Focus on your driving instead of reacting emotionally.

⛽ Take Breaks if Needed

Long drives can be tiring. If you feel stressed or tired, take a short break and relax.

⚠️ Final Tip:

Confidence comes from practice + patience. It's okay to feel nervous at first—ever experienced driver was once a beginner.

Observation:

Aspect	Naïve Prompt	Basic Prompt
Quality	Low to average – generic, uninspired	High – engaging, polished, creative
Accuracy	Generally correct but minimal	Accurate and complete
Depth	Shallow, surface-level	Detailed, thorough, insightful
Structure	Poor or none	Well-organized (clear flow, lists, etc.)
Length & Relevance	Often too short or slightly off-topic	Appropriate length, directly on target

Result:

Thus, the comparative analysis of different types of prompting patterns and explain with various test scenarios have been evaluated and presented in the observation table.